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Department of Mathematical and Physical Sciences, EWU
Midterm Examination, Spring-2026
Course: STA 102 (Statistics and Probability), S-10
Full Marks: 30, Time: 1 Hour & 10 Minutes
Answer all the questions. Figures in the right margin indicate marks.

1. A university conducted a study to analyze the coding practices and academic engagement of Computer Science and Engineering (CSE) students. Data were collected from 75 students through an online survey and academic records. Each student's data include the following variables:
   I. Enrollment Year (2024, 2025, etc)
   II. Gender (Male, Female)
   III. Average Daily Coding Time (in hours spent programming or practicing coding)
   IV. Number of Completed Programming Projects (personal or academic)
   V. Preferred Programming Language (Python, C++, Java, Others)
   VI. Satisfaction with University Internet Facilities (rated on a 5-point Likert scale: 1 = Very Dissatisfied to 5 = Very Satisfied)

   a) Identify the population and sample for the study. (3)
   b) Write down the appropriate scale of measurement for each variable. (3)

2. A network provider investigates the load of its network by determining the number of concurrent users from a random sample of fifty locations. The frequency distribution of concurrent users (in thousands of people) is given as follows.

Load	10.1-12	12.1-14	14.1-16	16.1-18	18.1-20	20.1-22	22.1-24
Frequency	2	4	8	13	11	7	5

   a) Draw an ogive and hence estimate what percentage of the place had load between 15 and 20? (4)
   b) Find the mean and the median of the load. (4)

3. A CSE department is evaluating the lifetime of laptop batteries (in hours) used by students in their lab. The variable is denoted by x. They collected data from 30 laptops over several semesters on that variable. The data (sorted in ascending order for convenience) are as follows:
   0.4 , 0.6 , 1.0 , 1.1 , 1.2 , 1.5 , 1.5 , 1.6, 2.1, 2.4, 2.7, 2.8, 3.1, 3.3, 3.5, 3.8, 3.9, 4.2, 4.4, 4.6, 4.7, 5.0, 5.4, 5.5, 5.7, 6.9, 7.1, 8.5, 8.9, 9.9
   From the data the following values have been computed $\sum x = 117.3$, and $\sum x^2 = 643.57$

   a) Find the standard deviation of the lifetime of laptop batteries. Compute the z-score for x = 1.35 and comment. (3)
   b) Given that the data does not contain outlier, draw a box plot and explain the skewness. (3)
   c) From another department, the mean and the standard deviation of the lifetime of laptop batteries were computed as 3.04 and 2.42, respectively. Explain which department's lifetime of laptop batteries is relatively stable. (2)

4. A study was conducted to examine whether internet connectivity influences the use of cloud-based coding platforms among Computer Science and Engineering (CSE) students. With the growing use of online programming environments, reliable internet access has become important for coding practice, project development, and collaboration. Data were collected from 12 CSE students. For each student, two variables were recorded: Internet Speed (in Mbps), x and Frequency of Using Cloud Coding Platforms (times per week), y. The following values were obtained from the data:
   $\sum xy = 1827, \sum x^2 = 7657, \sum x = 263, \sum y^2 = 448, \sum y = 68$.

   a) Compute the Pearson correlation coefficient r between the two variables. (2)
   b) Estimate the regression equation $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ and interpret the slope parameter. (3)
   c) Predict y for a student whose internet speed is 28 Mbps and explain the results. (2)
   d) Compute and interpret the coefficient of determination. (1)